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PureML: a transparent NumPy-only deep learning framework for teaching and prototyping

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Software

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Summary

PureML is a compact deep-learning framework implemented entirely in NumPy. It provides a tensor type with reverse-mode automatic differentiation, core neural-network layers and losses, optimizers and learning-rate schedulers, activations, training utilities, and persistence for model and optimizer states. A packaged MNIST dataset makes it easy to benchmark or teach end-to-end. The package targets learners, educators, and researchers who want to inspect and prototype deep-learning ideas with full control and transparency over the code involved, without heavy multi-language codebases that primarily optimize for production deployments and obscure the machine-learning theory underlying the implementation.

Statement of need

Modern deep-learning libraries such as PyTorch and JAX provide rich ecosystems but are conceptually and operationally heavy for teaching low-level ML theory, code reading, or CPU-only environments (Bradbury et al., 2018; Paszke et al., 2019). Educational materials often rely on pseudo-code or small snippets that omit practical details: broadcasting semantics, batching, parameter persistence, computational graph construction, vectorization, gradient accumulation, checkpointing, etc. As a result, learners struggle to bridge the gap to real systems. At the other end of the spectrum, educational projects like `micrograd` (Karpathy, 2020) purposefully keep the scope tiny (scalar or minimal tensor autodiff with a small MLP), which makes them excellent teaching tools but leaves out the aforementioned real-world challenges. Projects like `numpy-ml` (Bourgin, 2019) offer a broad catalog of algorithms implemented in NumPy but are not built around an autodiff engine and are not aimed at performance; they serve as references rather than frameworks for training deep networks.

PureML aims to sit between these extremes: small enough to audit end-to-end, but feature-complete enough for nontrivial models. The code remains transparent while supporting batch-vectorized computation, dynamic computational graphs, forward pass caching, persistence, and related functionality. It focuses on:

- Explicit reverse-mode autodiff with readable vector-Jacobian products for every operation, so gradient flow is inspectable.
- Minimal runtime dependencies (NumPy and zarr) suitable for laptops, classrooms, and CPU-only servers (Harris et al., 2020; Miles et al., 2025).
- Ready-to-run MNIST example to demonstrate end-to-end training without additional downloads (LeCun et al., 1998).
- Persistence utilities that round-trip models, optimizer slots, and data for reproducible exercises or small experiments.

Design and implementation

Core autograd. The `Tensor` type wraps NumPy arrays, records operations dynamically, and executes reverse-mode autodiff with explicit VJPs. Broadcasting-aware gradient reduction, slice/advanced indexing support, and graph teardown utilities (`zero_grad_graph`, `detach_graph`) mirror the behaviors found in larger frameworks while remaining short enough to audit. Safe exports (`Tensor.numpy`) discourage in-place mutation of parameter buffers.

Layers and losses. The library supplies `Affine`, `Dropout`, `BatchNorm1d`, and `Embedding` layers. Losses include mean squared error, binary cross-entropy (probabilities or logits), and categorical cross-entropy with optional label smoothing. Stable softmax and log-softmax implementations avoid overflow.

Optimization stack. Optimizers (SGD with momentum, AdaGrad, RMSProp, Adam/AdamW) share a common interface, support coupled or decoupled weight decay, and persist optimizer slots via `save_state`/`load_state`. Lightweight schedulers (step, exponential, cosine annealing) operate in-place on optimizer learning rates.

Data utilities and models. A `Dataset` protocol, `TensorDataset`, and `DataLoader` (with slicing fast paths and optional shuffling) simplify input pipelines. The bundled `MnistDataset` streams compressed images/labels from a packaged zarr archive (LeCun et al., 1998). Example models include a small fully connected MNIST classifier (`MNIST_BEATER`) and a classical k-nearest neighbors classifier.

Persistence. The `ArrayStorage` abstraction wraps zarr v3 groups with Blosc compression and can compress to read-only zip archives (Miles et al., 2025). Model parameters, buffers, and top-level literals can be round-tripped to a single `.pureml.zip` file for reproducibility.

Ecosystem and dependencies. PureML requires only NumPy and zarr at runtime (Harris et al., 2020; Miles et al., 2025), targets Python 3.11+, and is distributed on PyPI as `ym-pure-ml`. Logging utilities configure rotating file/console handlers for experiments.

Project structure at a glance (code modules):

```
pureml/
  machinery.py        # Tensor core, autograd graph/VJPs
  layers.py           # Affine, BatchNorm1d, Dropout, Embedding
  losses.py           # CCE, BCE, MSE
  activations.py      # relu, softmax, log-softmax, etc.
  optimizers.py       # SGD, Adam/AdamW, RMSProp, AdaGrad + schedulers
  training_utils.py   # DataLoader, batching/loop helpers
  datasets/
    MNIST/dataset.py # packaged MNIST reader (zarr)
  models/
    neural_networks/mnist_beater.py
    classical/knn.py
  util.py             # ArrayStorage (zarr persistence), helpers
  base.py             # NN base class (save/load, train/eval)
  evaluation.py       # metrics (accuracy)
  general_math.py     # math helpers
  logging_util.py     # logging setup
```

Quality control

The GitHub repository contains a unit test suite (`tests/`) consisting of 106 tests that cover autograd correctness (elementwise ops, broadcasting, slicing, matmul, reshaping), activation stability, layers and buffers (including bias toggles and seeding), optimizer and scheduler

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behavior, persistence round-trips, data utilities, and the MNIST dataset/model flow. The suite runs with `python -m unittest discover tests`.

Example usage

```python
from pureml import Tensor
from pureml.activations import relu
from pureml.layers import Affine
from pureml.base import NN
from pureml.datasets import MnistDataset
from pureml.optimizers import Adam
from pureml.losses import CCE
from pureml.training_utils import DataLoader
from pureml.evaluation import accuracy
import time

class MNIST_BEATER(NN):

    def __init__(self) -> None:
        self.L1 = Affine(28*28, 256)
        self.L2 = Affine(256, 10)

    def predict(self, x: Tensor) -> Tensor:
        x = x.flatten(sample_ndim=2) # passing 2 because imgs in MNIST are 2D
        x = relu(self.L1(x))
        x = self.L2(x)
        if self.training:
            return x
        return x.argmax(axis=x.ndim-1) # argmax over the feature dim

with MnistDataset("train") as train, MnistDataset("test") as test:
    model = MNIST_BEATER().train()
    opt = Adam(model.parameters, lr=1e-3, weight_decay=1e-2)
    start_time = time.perf_counter()
    for _ in range(5):
        for X, Y in DataLoader(train, batch_size=128, shuffle=True):
            opt.zero_grad()
            logits = model(X)
            loss = CCE(Y, logits, from_logits=True)
            loss.backward()
            opt.step()
    end_time = time.perf_counter()
    model.eval()
    acc = accuracy(model, test, batch_size=1024)
print("Time taken: ", end_time - start_time, " sec.")
print(f"Test accuracy: {acc * 100}")
```

Example usage in computational biology

SAWNERGY project builds its skip-gram embedding pipeline for amino acid interaction networks using PureML (link).

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Availability

Source code: https://github.com/Yehor-Mishchyriak/PureML
PyPI: https://pypi.org/project/ym-pure-ml/
Documentation: https://ymishchyriak.com/docs/PUREML-DOCS
License: Apache-2.0 (see `LICENSE`)

Future directions

Planned extensions include convolutional, recurrent, and message-passing layers, attention mechanisms, additional activation and loss functions, richer evaluation metrics, and related tooling to support a broader range of deep-learning experiments.

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